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Simon Allerton

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Vertical-axis rotation associated with folding and thrusting: An example from the eastern Subbetic zone of southern Spain

Simon Allerton
Department of Earth Science, University of Oxford, Parks Road, Oxford OX1 3PR, United Kingdom

ABSTRACT

The differential rotation about a vertical axis across a southeast-vergent recumbent fold and thrust fault within a single nappe from the eastern Subbetic zone of southern Spain has been determined using paleomagnetism. The upper limb has rotated clockwise by 68° ± 22° relative to the lower limb. Mapping of the structure indicates that the syncline is an isoclinal fold in the northeast and an open fold in the southwest. The differential vertical-axis rotation across this structure appears to have been accommodated by a progressive increase in along-strike horizontal shortening, reaching 8.5 km at 3.7 km from the pivot point. This differential shortening and the rotation are probably the result of increased drag on the underlying thrust close to the pivot point compared to the farthest extent of the structure. The lateral boundary of the rotating sheet is a sinistral strike-slip fault, which is consistent with the rotational geometry. Pinning is probably enhanced by lateral heterogeneities within the stratigraphy. This result ties rotational deformation to early-middle Miocene folding and thrusting, rather than to late Miocene–Holocene strike-slip deformation. Rotation by pinning of thrust sheets may prove to be an important aspect of deformation in many fold and thrust belts.

INTRODUCTION

Since fault-block rotations about a vertical axis were first identified in the western United States (e.g., Beck, 1976; Luyendyk et al, 1980), they have been observed in many tectonic environments. They are now recognized to represent an important component of deformation.

Models that describe the rotation of fault blocks can be divided into two fundamentally different types. The first group regards rotations as responses to boundary conditions of the deforming zone, and assumes that a brittle layer responds to flow in the ductile layer, which can be modeled by fluid dynamics (e.g., McKenzie and Jackson, 1983). The second group relates block rotations to the geometry of the fault systems that bound them. In these models the blocks act as rigid sheets that deform because of stresses at their boundaries (e.g., Garfunkel, 1974). The two types of models are not irreconcilable: the gross pattern of rotation may be controlled by boundary conditions of the deforming zone, and local variations in rotation may result from local fault geometries. Many of these models assume that the blocks are on a crustal scale and that the faults cut the entire upper crust (e.g., Garfunkel, 1974; McKenzie and Jackson, 1983). They are most appropriate for describing deformation in strike-slip systems. In addition to these thick-skinned situations, paleomagnetic rotations associated with the transport of large-scale nappes have been described (e.g., Kissel et al., 1993).

None of these models can easily account for large, differential rotations, as have recently been described in the eastern Subbetic zone of southern Spain (Allerton et al., 1993), where rotations >60° between individual fault blocks have been described. This paper focuses on a single thrust sheet from which Allerton et al. (1993) published data from two sites with a large relative rotation between them. I have used paleomagnetism to investigate this rotation further, and I have mapped the structure in detail to determine how this rotation was accommodated.

GEOLOGIC SETTING

The Betic Cordillera of southern Spain (Fig. 1) is part of the Gibraltar arc, the westernmost extension of the Alpine orogen. It has been separated into the dominantly metamorphic Internal Zones, and the unmetamorphosed External Zones (Egeler, 1964). The External Zones are subdivided into the Subbetic and Prebetic zones, consisting, respectively, of basinal and shelf facies Mesozoic and Tertiary sedimentary rocks that were thrust northwestward onto the Iberian platform during Miocene time. Early-middle Miocene folding and thrusting of the Subbetic zone was followed by strike-slip deformation from late Miocene time, synchronous with continued thrusting in the Prebetic zone (Platt and Vissers, 1989).

Paleomagnetic work in the western Betic Cordillera has documented a consistent pattern of clockwise rotations (Osete et al., 1989; Platzman, 1992). In the eastern Subbetic zone, a general clockwise rotation has been observed (Allerton et al., 1993), but individual thrust sheets have rotated by different amounts, producing a much more complicated pattern of rotation.

STRUCTURE AND STRATIGRAPHY

The Quipar-Burete area can be divided into an upper and lower thrust sheet (Figs. 1, 2, and 3). The upper sheet crops out in the Sierra de Quipar and the Sierra de Burete. The lower sheet is exposed in a structural window between the Sierra de Quipar and the Sierra de Burete, and to the southeast of the Sierra de Burete. The principal difference between the two is in the Upper Jurassic sequence. In the upper sheet the Upper Jurassic is represented by a sequence of well-bedded limestones with cherts and up to 100 m of Ammonitico Rosso facies limestones. The Upper Jurassic of the lower sheet is a sequence of well-bedded limestones with calpionellids, and the Ammonitico Rosso facies limestones are absent. The massive Lower Jurassic carbonates are locally absent from the lower sheet, and where this happens, the Upper Jurassic

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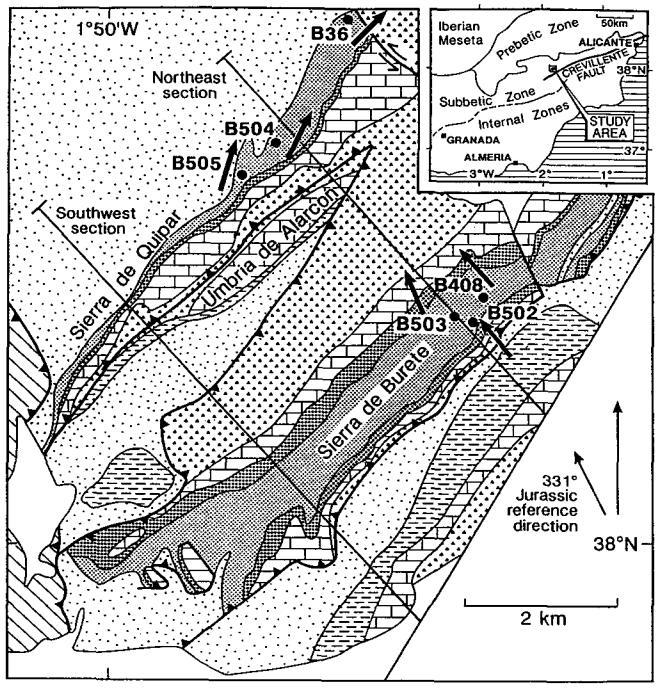


Figure 1. Geologic map of study area within eastern Subbetic. See Figure 2 for key to patterns. Diagonal rule represents higher level nappe. Thick arrows indicate mean site paleomagnetic declinations for high-temperature components (after tectonic correction). Two northwest-southeast lines show locations of sections illustrated in Figure 3. Large solid circles indicate sampling sites. Inset: Southeast Spain showing location of study area.

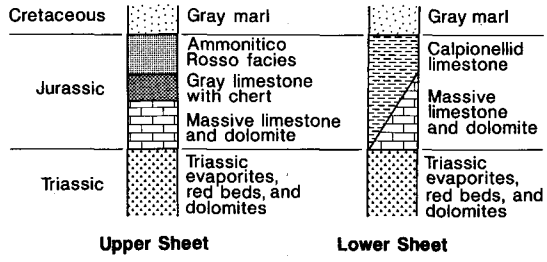


Figure 2. Stratigraphy of Quipar-Burete area.

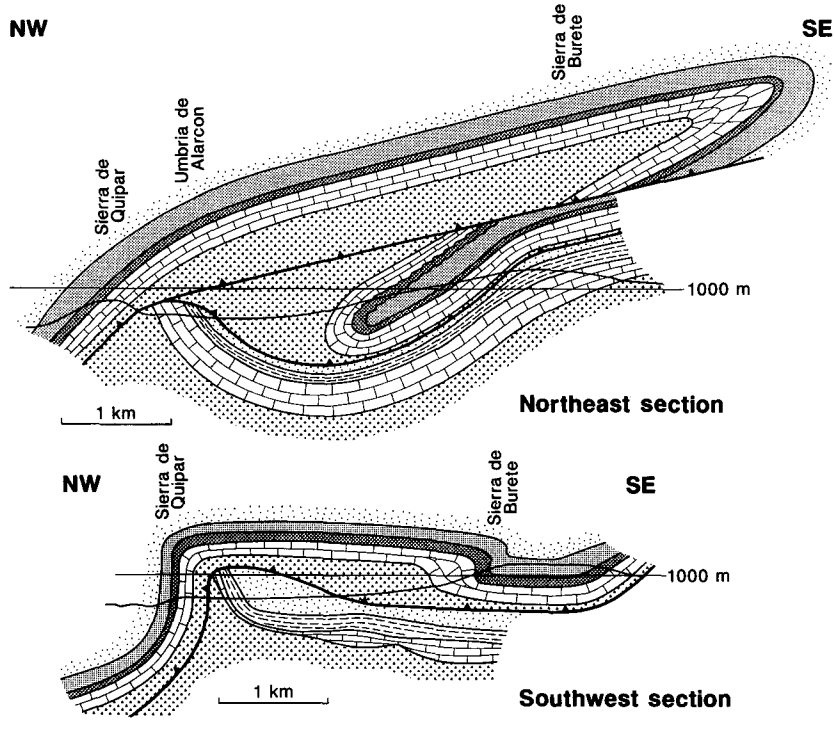


Figure 3. Structural sections (see Fig. 1 for locations and Fig. 2 for key). No vertical exaggeration.

limestones are directly on top of the Triassic evaporitic sequence.

The stratigraphic sequence illustrated in Figure 2 makes a sandwich of contrasting rheologies. At the base is a Triassic sequence composed of highly tectonized evaporites, red beds, and dolomites. Above this are Jurassic carbonates, overlain by Lower Cretaceous marlstone. This produces a characteristic deformation style; the Jurassic carbonates act as a competent rigid sheet and the Triassic and Cretaceous rocks deform in a more incompetent manner.

The upper sheet has been thrust over the lower thrust sheet and folded into a southeast-vergent recumbent fold that is tight to isoclinal in the northeast section, and open to closed in the southwest section (see Fig. 3). In this structural framework the Sierra de Quipar formed as a homocline above a footwall ramp, the Sierra de Burete formed as a recumbent syncline, and the Umbria de Alarcon formed as an oversteepened footwall ramp. In the north of the Sierra de Burete there is an extensive outcrop of the overturned limb of the syncline, and associated parasitic folds. Lower Cretaceous rocks are absent from the core of the syncline. In the south, the fold is exposed on the northwest flank of the Sierra de Burete as an open to closed structure with a short, vertical central limb. Lower Cretaceous rocks crop out in the core farther to the southwest.

I have drawn the balanced section assuming that the anticline has a geometry similar to that of the syncline and that the recumbent fold has been breached by thrusting through the overturned limb. Note that there is considerably more shortening in the northeast of the Sierra de Burete fold structure than in the southwest. This differential shortening must be accommodated either by lateral ramps (for which there is no evidence) or by a significant vertical-axis rotation.

PALEOMAGNETIC SURVEY

The relative vertical-axis rotation of the upper sheet between the Sierra de Quipar and the Sierra de Burete has been determined by a paleomagnetic survey. Previous work in the area (Allerton et al., 1993) demonstrated that the Upper Jurassic Ammonitico Rosso facies carries a stable remanence that predates middle Miocene folding and passes a conglomerate test, indicating a pre-Oligocene age for the remanence. It is thus ideally suited to identifying tectonic rotations associated with the middle Miocene shortening event. Rock magnetic experiments (Allerton et al., 1993) suggest that the carrier is dominantly magnetite.

In addition to two sites from the area studied by Allerton et al. (1993), we have sampled two sites from the Quipar homo-

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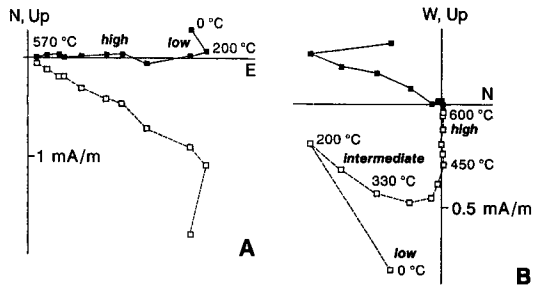


Figure 4. Orthogonal vector diagrams of typical stepwise thermal demagnetization behavior for Upper Jurassic Ammonitico Rosso facies samples (in situ; open symbols = points in vertical plane; solid symbols = points in horizontal plane). A: Example of two-component behavior from sample B504-11b from Sierra de Quipar. B: Example of three-component behavior from sample B502-4b from Sierra de Burete.

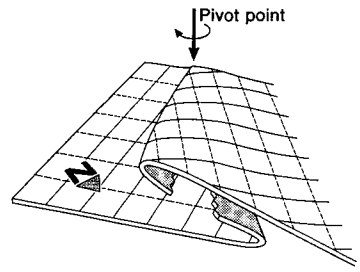


Figure 5. Cartoon illustrating rotation associated with differential shortening related to folding and thrusting of Sierra de Burete and Sierra de Quipar. Approximately 70° of clockwise rotation is accommodated by 8.5 km of shortening at 3.7 km from pivot point.

cline and two from the lower limb of the Burete recumbent syncline. The sites on the Sierra de Quipar yield two components (see Fig. 4A) on thermal demagnetization. The low-temperature (T) component (<300 °C) is close to the present field direction, before tectonic correction. The high-T component (300–570 °C) includes sites with both normal and reversed polarities. The scatter of sites for this component is marginally less, although not significantly so, after a correction for bedding has been applied.

The sites on the lower limb of the Sierra de Burete syncline have three components (see Fig. 4B). One site (B408, Table 1) was sampled across an outcrop-scale fold; this allowed the relative age of these components to be identified. The low-T component (<200 °C) postdates folding and is close to the present field direction. The intermediate-T component (200 to 420 °C) similarly postdates folding, but represents a reversed polarity direction. The high-T component (420 to 570 °C; Table 1) predates folding and includes normal and reversely magnetized components. Similar components have been identified in the other sites.

The low-T component probably represents a recent overprint. The absence of the intermediate-T component from the rocks of the Sierra de Quipar, which have only undergone a minor degree of tectonism, and its presence in those of the tightly folded Sierra de Burete structure suggests that this component probably represents a late tectonic (middle Miocene) overprint. The high-T component predates folding, contains both normal and reverse polarities, and is comparable to the high-T component obtained for the Sierra de Quipar. The high-T component can therefore be used to determine any between-site rotations.

GEOMETRY OF ROTATION

The site mean directions for the high-T component (see Table 1 and Fig. 1) within each limb are clearly similar. The sites from the Sierra de Quipar have a mean direction, after bedding correction, of declination 031°, inclination 56°, and α_{95} = 16°; those from the Sierra de Burete have a direction of declination 323°, inclination 35°, and α_{95} = 15°. The declination anomaly, compared to an Iberian Jurassic reference (331°; Allerton et al., 1993) is −8° ± 15° for the Sierra de Burete, and 60° ± 16° for the Sierra de Quipar, and thus the upper limb of the fold has rotated 68° ± 22° clockwise relative to the underlying Burete limb.

On first inspection this appears to be a very large relative rotation, but because of the small scale of the structure involved, this large rotation can be accommodated by a reasonable amount of differential shortening. The two structural sections shown in Figure 3 are separated by about 3 km along strike. A relative rotation of 68° can be generated by 7 km greater shortening in the north than in the south. This geometry has been incorporated into the construction of the balance sections, so that the southwestern section accommodates about 1.5 km of shortening and the northeastern section about 8.5 km. The pivot point lies about 0.7 km southwest of the southwestern section (see Fig. 5). The eastern boundary of the Sierra de Quipar is a north-northwest–trending sinistral strike-slip fault, which apparently has some dip-slip component because it juxtaposed Triassic rocks northeast of the fault against the Jurassic rocks of the Sierra de Quipar. The geometry of this fault is entirely consistent with the rotational folding and thrusting of the upper sheet, because it acts as a transfer fault. To the west, the upper sheet of the Sierra de Quipar is overthrust by a higher level nappe.

The strike of the Sierra de Quipar is approximately parallel to the axial trace of the Burete syncline, despite the large relative rotation between them, because the orientation of the Sierra de Quipar is controlled by the ramp in the footwall, which has not been rotated relative to the Burete syncline.

For rotation to occur, a shear couple must be generated across the upper sheet. A simple way of doing this is by increasing drag ("pinning") close to the pivot point of the structure. Supporting evidence comes from inspection of the structural section (Fig. 3). The footwall ramp in the southwestern section is steeper, and reaches a greater elevation than the ramp in the northeastern section; it would therefore represent a greater barrier to an overriding thrust sheet.

DISCUSSION

The results of this study have some important implications for the tectonic evolution of the eastern Subbetic zone. I have

TABLE 1. PALEOMAGNETIC RESULTS FOR HIGH-TEMPERATURE COMPONENTS

Site	Before correction						After correction				Bedding	
	Dec.	Inc.	k	α95	N	n	Dec.	Inc.	k	α95	Dip	Direction
Sierra de Quipar												
B36	062	46	50.8	7.3	9	9	046	50	50.8	7.3	15	310
B504	094	34	58.2	5.7	12	12	027	62	58.2	5.7	57	308
B505	069	51	142.3	4.3	10	9	017	54	142.3	4.3	37	309
Mean	076	45	28.4	23.6	3	3	031	56	58.5	16.3		
Sierra de Burete												
B408	307	35	11.6	18.5	11	6	316	29	32.1	10.8		
B502	243	81	42.3	6.7	12	12	319	37	42.3	6.7	25	350
B503	341	79	12.4	16.5	9	9	336	37	12.4	16.5	42	334
Mean	305	68	8.0	46.8	3	3	323	35	66.0	15.3		

Note: Precision parameters k and α95 are Fisher's dispersion parameter and 95% cone of confidence, respectively. N is the number of demagnetized samples, n is the number used in the statistics. Bedding is specified as dip and direction of dip. No bedding measurements are given for site B408, which was sampled across a fold. Stable component for site B503 was isolated by the great circle technique.

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developed a structural model involving differential shortening as a mechanism for generating large, differential rotations that may also be applied to some of the previously published paleomagnetic results (e.g., Allerton et al., 1993). Linking rotation to thrusting also constrains the timing of rotational deformation to early-middle Miocene time. Some authors (e.g., Osete et al., 1989) have speculated that the rotation was associated with strike-slip movement on the Crevillente fault (which is about 5 km south of the Sierra de Burete) during late Miocene–Holocene time. It is difficult to identify a strike-slip faulting model that can account for a large (68°) rotation, but that would not produce a similar rotation of the earlier fold structures (i.e., Sierra de Quipar and Sierra de Burete). The structural evidence (Banks and Warburton, 1991; Allerton et al., 1994) indicates that thrusts are relatively thin skinned and are not shallow slivers off a deeper fault structure (De Smet, 1984).

Kinematic data (Frizon de Lamotte et al., 1989; Lonergan et al., 1993; Allerton et al., 1994) suggest that thrust transport directions within the Subbetic zone were highly oblique to its margins. The general clockwise rotation of the thrust sheets within the Subbetic zone is consistent with this sense of oblique convergence.

The high degree of variability of rotation is not a consequence of the overall large-scale kinematics of the orogen. Thrust-sheet pinning provides a simple mechanism for differential rotations. This effect is probably enhanced by the highly variable stratigraphy observed in the eastern Subbetic zone. In the western Subbetic zone the Lower Jurassic exists across the area as a continuous sheet of shelf carbonates, and rotational deformation is very consistent; rotations are between 30° and 60° (Osete et al., 1989). In comparison, the eastern Subbetic zone exhibits a highly disrupted Jurassic stratigraphy, particularly in the center of the Subbetic zone, where Middle to Late Jurassic rifting has significantly dissected the Lower Jurassic shelf carbonates. In some places, including the lower thrust sheet, Upper Jurassic and occasionally Lower Cretaceous rocks have been deposited directly onto Triassic rocks. This produces a highly heterogeneous rheology in place before shortening commenced in early-middle Miocene time, which in turn could have resulted in the pattern of highly variable rotations observed in the eastern Subbetic zone.

These results also have implications for other fold belts. If differential rotations are controlled by lateral heterogeneities in the stratigraphy, rather than by the overall geometry and kinematics of the orogen, they may be expected in any tectonic environment with a similar stratigraphy. They may be widespread where a shelf sequence has been extended and then shortened, a common occurrence in inverted marginal basins. Parallelism of structures that are controlled by the geometry of the footwall is apparently no assurance that there has been no relative rotation within the hanging wall.

The structure illustrated in this study may also be considered a salutary warning for paleomagnetists. First inspection of the Quipar-Burete nappe would suggest that it is an ideal candidate for a regional-scale fold appropriate for a paleomagnetic fold test. However, the noncylindricity of this fold is not obvious from the geologic map (Instituto Geológico y Minero de España, 1974), and detailed mapping was required to define the rotational nature of this structure.

CONCLUSIONS

Paleomagnetic evidence suggests that there has been a 68° ± 22° clockwise vertical axis rotation of the upper limb (Sierra de Quipar) of a recumbent fold relative to the lower limb (Sierra de Burete). Mapping of the lower syncline indicates that it is isoclinal in the north and open in the south. This structure can accommodate ~70° of rotation if it has involved about 8.5 km more shortening in the northeast than at the pinpoint 3.7 km to the southwest. This result links rotation to early-middle Miocene folding and thrusting. Differential drag on the basal thrust caused by variable footwall geometry is a plausible mechanism for generating the large differential rotations described here and elsewhere in the eastern Subbetic zone (e.g., Allerton et al., 1993). Heterogeneity of the Jurassic carbonate stratigraphy may play an important role in pinning thrust sheets, causing rotation. The overall geometry of the orogen probably controls the large-scale rotational pattern, but not the local, differential effects.

This mechanism for rotation by thrust faulting and folding suggests that vertical-axis rotational deformation in fold belts may be more widespread than previously suspected. Such rotations may be common where a heterogeneous stratigraphy is involved.

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